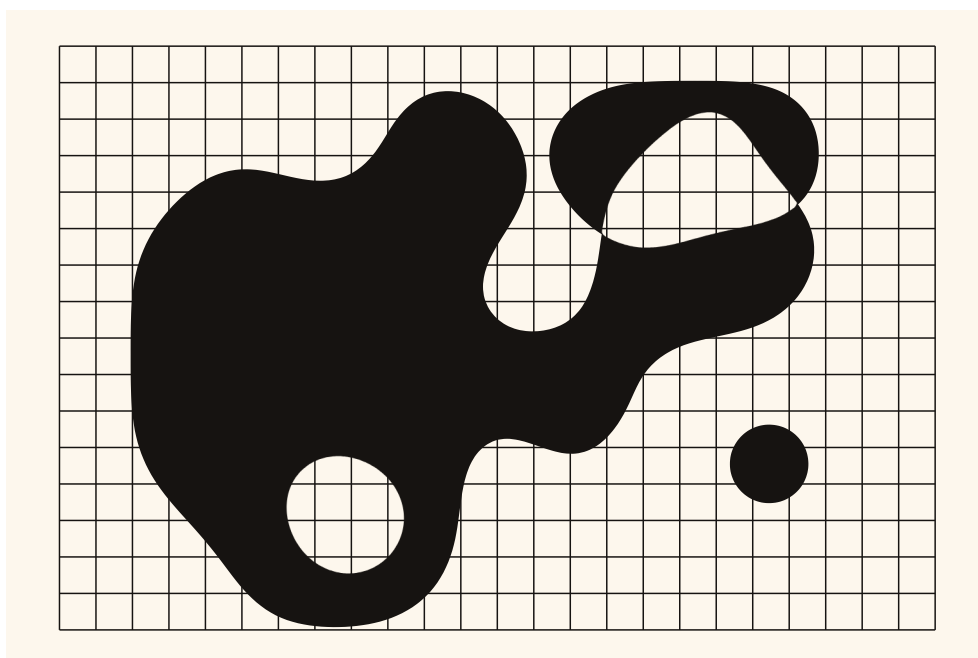


Inkling: наша модель с открытыми весами

15 июля 2026 г.

[Попробовать Tinker](#)[Карточка модели >](#)[Hugging Face >](#)

Наша миссия — создать искусственный интеллект, который расширит возможности человеческой воли и разума. Мы разработали платформу, которая позволяет любому пользователю настраивать модели, представили систему искусственного интеллекта, созданную для интерактивного сотрудничества, и опубликовали новое исследование. Сегодня мы продвигаемся к нашей цели, выпуская модель, которую мы обучили с нуля, с полным набором весов, чтобы люди могли адаптировать ее под себя.

Наша модель под названием Inkling представляет собой трансформер Mixture-of-Experts с общим количеством параметров 975 млрд, из которых 41 млрд — активные. Она поддерживает контекстное окно до 1 млн токенов. Модель была предварительно обучена на 45 триллионах токенов текста, изображений, аудио и видео. Это первая модель в семействе моделей разных размеров. Вместе с ней мы представляем предварительную версию Inkling-Small — более легкой модели с 12 миллиардами активных параметров, обученной по аналогичной методике. Она обеспечивает высокую производительность при еще более низкой стоимости и задержке.

Inkling изначально обрабатывает текст, изображения и аудио и обеспечивает баланс между стоимостью и производительностью за счет эффективного и контролируемого мышления. Мы обучили ее как универсальную,

доступных на сегодняшний день моделей, как открытых, так и закрытых. Однако сочетание ее качеств делает ее хорошей основой для настройки: мультимодальные возможности, эффективное мышление и доступность на Tinker для тонкой настройки. Inkling — это только начало: наш первый релиз в семействе моделей, которое мы продолжим развивать.

Мы хотим сделать кастомизацию доступной для большего числа сценариев использования, поэтому Inkling уже сегодня доступен для тонкой настройки на Tinker. Выбор подходящей базовой модели для тонкой настройки — это субъективная оценка, которая сочетает в себе измеримые показатели и уникальные ощущения от взаимодействия с моделью. Чтобы упростить последнее, мы добавили Inkling Playground в консоль Tinker: интерфейс для разработчиков, позволяющий взаимодействовать с Inkling.

Чтобы показать, что такое кастомизация на практике, мы попросили Inkling провести тонкую настройку. С помощью Tinker модель написала собственный скрипт для тонкой настройки, запустила его и оценила результат:

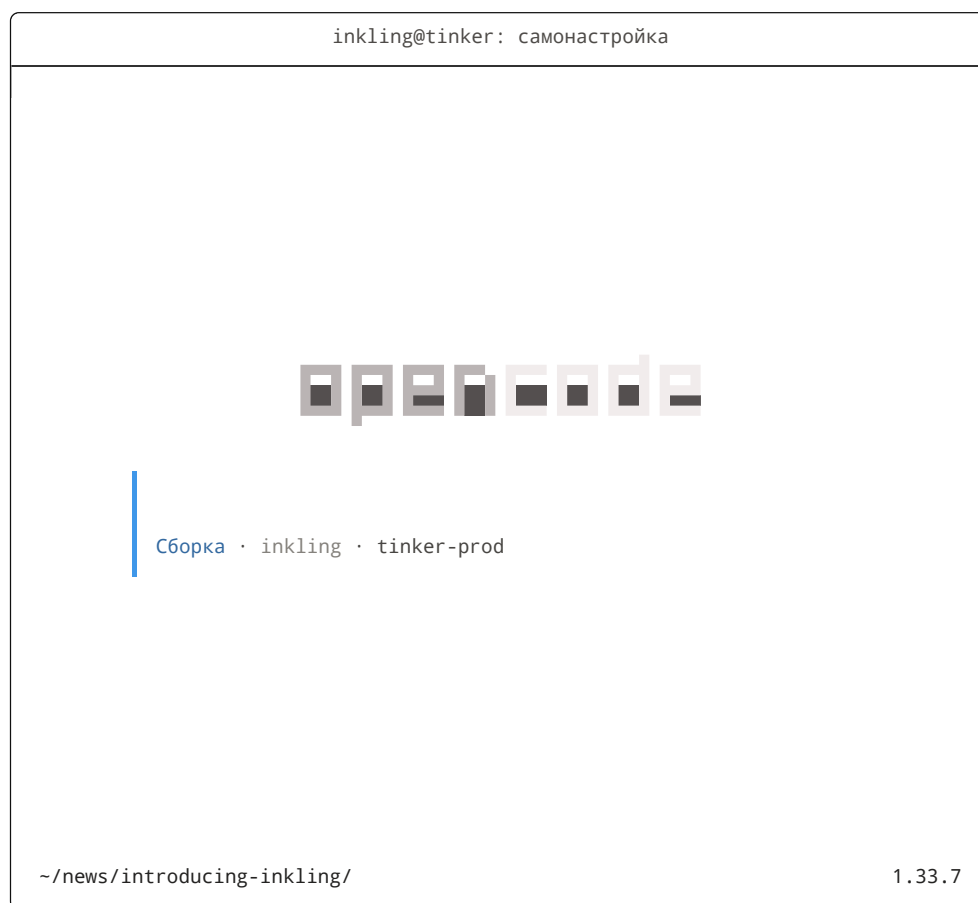


Figure 1 - start in OpenCode: Inkling runs inside the OpenCode harness.

Capabilities

it measures up on important qualities such as trustworthiness and safety.

Generalist model

Inkling is designed to be broad. We trained it across agentic, reasoning, coding, instruction-following, factuality, vision, and audio tasks, rather than narrowly optimizing for one domain. That breadth matters for customization and real-world use: different users need models that can adapt to very different workflows, not just excel on benchmarks.



Figure 2: Inkling is a broad, balanced generalist model. Benchmark scores are shown on a shared 0–100 scale; higher is better. The results show competitive performance across text, agentic, multimodal, and audio evaluations, rather than a model narrowly optimized for one benchmark family. This breadth reflects Inkling’s intended role: a practical multimodal foundation model for customization across domains, workflows, and products.

Agentic coding and tool use

A strong base for fine-tuning needs to flexibly solve a wide variety of tasks with agentic tool use. Inkling scores well among open-weights models on most agentic benchmarks.

We trained Inkling to run inside a variety of coding and agent harnesses, and we randomized the tool set and schema during training to reduce sensitivity to any particular one. Inkling’s controllable thinking effort, described in the next section, can be set from within the harness.

Below are a few demos showcasing Inkling’s agentic coding and tool use and the artifacts it creates.

making built a functional web app in a single shot, then powers an embedded AI assistant that can operate the web app interface through natural language instructions.

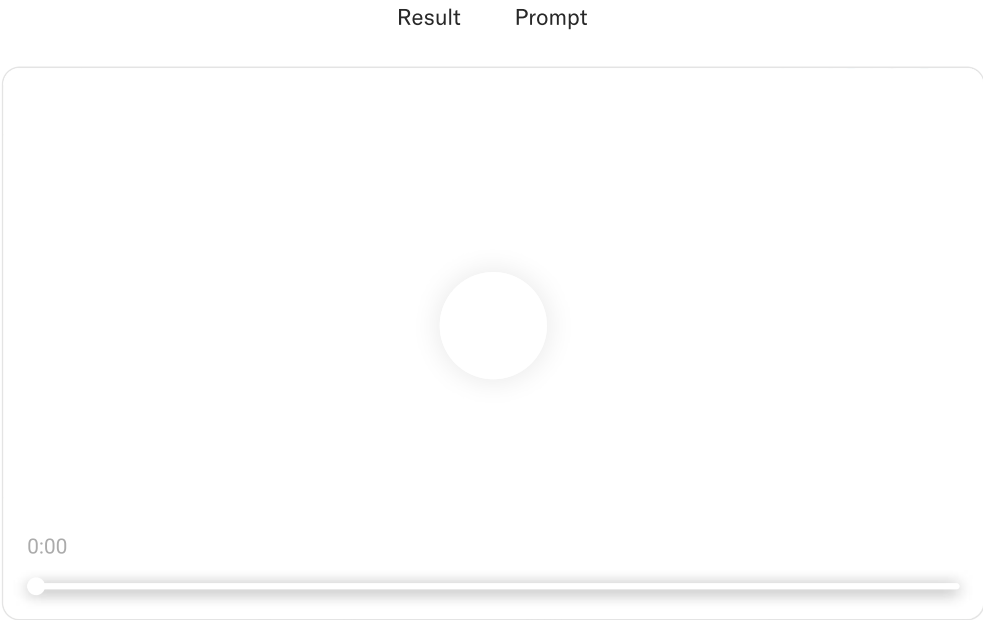


Figure 3: Inkling one-shots a job-application web app from the prompt in the second tab, then a browser-use agent fills out the form from a saved profile.

Design Arena

Inkling was evaluated on Design Arena’s Agentic Web Dev leaderboard, where blinded human evaluators compare generated web apps head to head. It ranks among the strongest open-weights models.

Claude Sonnet 5	1333
Claude Fable 5	1329
Claude Opus 4.8	1285
GLM 5.2	1275
Grok 4.5	1271
GPT-5.6 Sol	1260
Inkling	1257
Claude Opus 4.6	1257
Gemini 3.5 Flash	1254
Kimi K2.6	1249
Claude Sonnet 4.6	1237
Kimi K2.7 Code	1234
GLM 5.1	1233
Claude Opus 4.5	1212
Grok 4.20 Reasoning	1203
Gemini 3.1 Pro Preview	1187
Grok 4.3	1185
Kimi K2.5 (Thinking)	1185

Cohesively styled artifacts

Inkling creates multi-page artifacts with precise instruction following, accurate information, and cohesive styling and design throughout.

Result Prompt

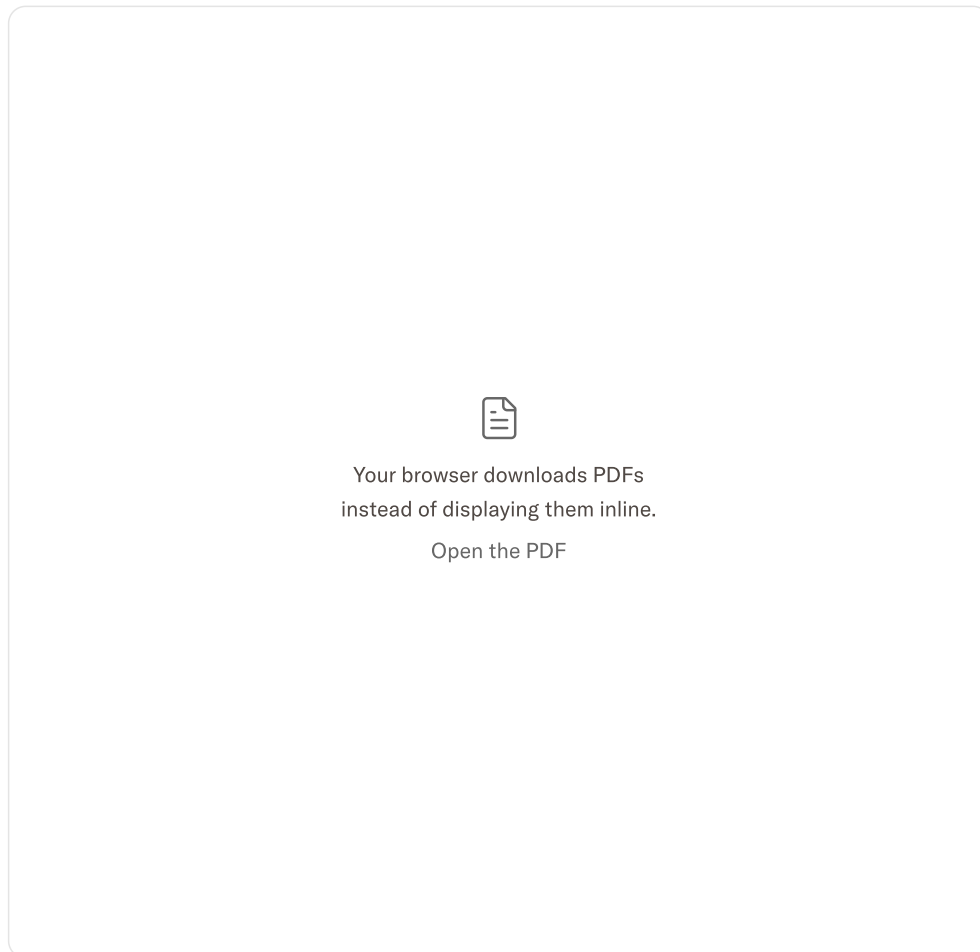


Figure 5: From the prompt in the second tab, Inkling produces a polished nine-page PDF food and travel journal — browse the actual document above.

Multiplayer game created through long refinement loop

Inkling refined an online snake game through 40 iterations of feedback from GPT Codex serving as a reviewer. The ability to sustain a long process of refinement and improve from feedback is crucial to creating the best collaborative work.

Result Prompt

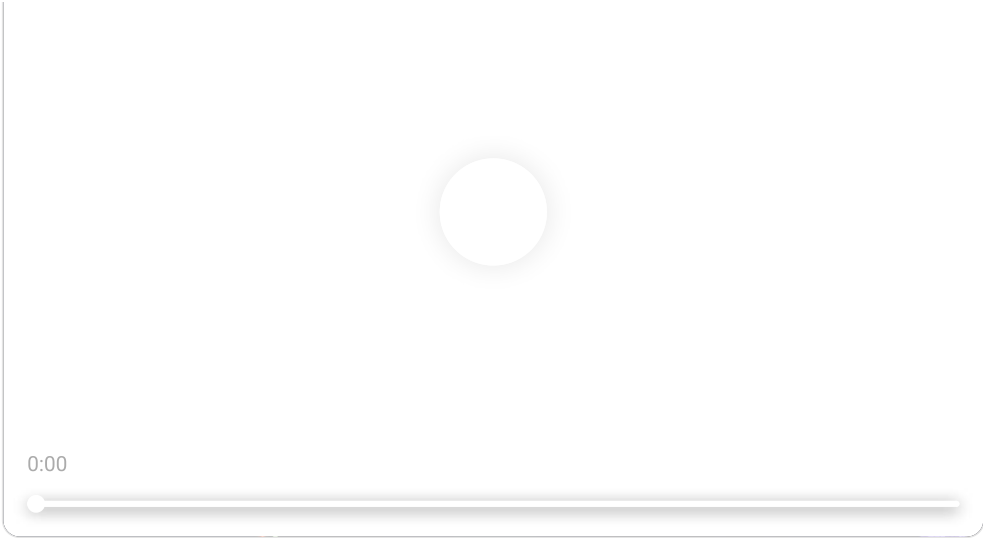


Figure 6: A multiplayer snake game generated by Inkling from the prompt in the second tab — real-time server, bots, leaderboard and all.

Controllable thinking effort

Test-time scaling and problem-solving are the core capability of every model, but that capacity is hard to capture with a single number. Developers fine-tuning models for a specialized task care as much about efficiency as about the max-effort performance on a public benchmark. Cost and latency are often binding constraints in real-world applications, and low latency in particular is crucial for enabling collaboration and improvement through iteration.

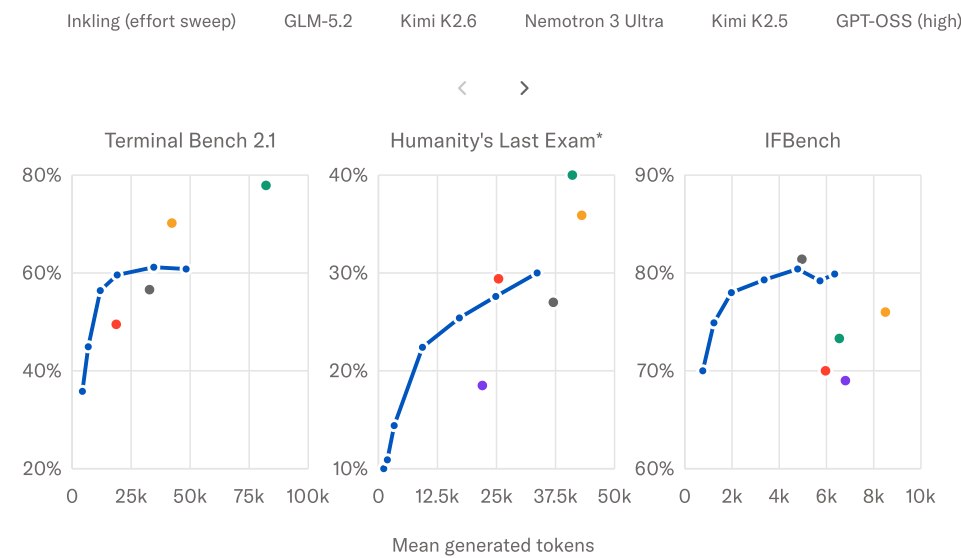


Figure 7: Sweeping Inkling's effort setting from 0.2 to 0.99 traces its performance against mean generated tokens on Terminal Bench 2.1, HLE, and IFBench; competing models are shown at their default operating point. Inkling reaches a given score at fewer tokens — for example, it matches Nemotron 3 Ultra on Terminal Bench 2.1 at roughly a third of the tokens. *Humanity's Last Exam scores reflect an earlier checkpoint and run slightly below the final release.

Inkling supports controllable thinking effort, allowing you to balance performance with token efficiency. The chart above shows the effort/performance curve of Inkling as well

Inkling spends one third as many tokens to achieve the same performance as Nemotron 3 Ultra on Terminal Bench. Cost and latency matter for a model that you run millions of times and as part of longer workflows; looking at the full cost curve allows developers to choose the best model for each use case.

Multimodality

A major goal of Inkling’s design is to serve as the background reasoning model in the [interaction models system](#) we recently introduced. Interaction models enable the user to collaborate naturally, using voice and vision in real time. This requires a model natively trained for broad multimodal capabilities.

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	Inkling effort=0.99	Open weights			
		Qwen3-Omni	Nemotron-3 Nano-Omni	Kimi K2.5	Kimi K2.6
Audio					
Audio MC	56.6%	24.3%	23.2%	–	–
MMAU	77.2%	77.5%	76.7%	–	–
VoiceBench	91.4%	88.8%	89.4%	–	–
Vision					
MMMU Pro (Standard 10)	73.5%	60.0%	53.0%	75.0%	79.0%
Charxiv RQ	78.1%	61.1%	63.6%	77.5%	80.4%
Charxiv RQ with python	82.0%	–	–	78.7%	86.7%

Audio and vision benchmarks against specialist omni models (open- and closed-weight), reported at effort=0.99.

The multimodal components were trained from scratch on general-domain data. We opted for an encoder-free architecture for audio and vision inputs, consistent with the interaction model design. Audio signals are input as dMel spectrograms¹, while images are encoded as patches of 40x40 pixels using a four-layer hMLP². Both are transformed via a light-weight embedding layer and processed jointly with text tokens.

Inkling transcribes speech, follows spoken instructions, answers questions about recordings, and reasons over longer-form audio. These capabilities place it among the strongest open-weights audio models on VoiceBench, MMAU, and AudioMC. For vision, Inkling accepts images as input and can describe visual content, answer questions, and perform in-depth reasoning based on the provided visual information. It demonstrates strong performance on charts, diagrams, and mathematical visual reasoning tasks. During inference, Inkling can also leverage a Python tool to support image understanding through operations such as zooming and cropping, while seamlessly integrating visual reasoning with code-based reasoning.

model and training pipeline in subsequent iterations.

Epistemics

We trained Inkling for calibration, instruction following, and resistance to censorship, which we refer to collectively as the model’s *epistemics*.

Getting the facts right requires more than memorizing a large corpus of knowledge. A useful model must be well-calibrated, expressing the right amount of confidence in its answers — including on questions which aren’t yet settled. The latter is a crucial capability for prediction and forecasting, an important use case where [fine-tuned models have shown rapid improvement](#) in recent months, [outperforming frontier LLMs](#).

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	Inkling	GPT-5.5	Claude Opus 4.8	Gemini 3.1 Pro	Grok
ForecastBench Brier Index † · no search	61.1 ± 0.79	59.1 ± 0.41	54.6 ± 0.99	61.1 ± 0.56	61.7 ±
ForecastBench Brier Index † · with search	63.7 ± 0.82	64.7 ± 1.28	58.6 ± 0.66	64.3 ± 1.05	63.2 ±
Prophet Arena Brier Score †	0.1617	0.1598	0.1605	0.1594	0.17

Results were obtained during testing between June 30 and July 13, 2026 on a different checkpoint of Inkling than the one released.

Forecasting requires integrating multiple sources of information into a calibrated probability, a core skill for a model users can trust. A model that’s confident in every answer it gives, including when it’s missing info and confabulates, forces the user to double-check everything. A model that gives the appropriate measure of confidence is useful across more real-world domains where information is often conflicting, unreliable, or hard to find. We trained for calibration with RL against proper scoring rules on a large corpus of resolved real-world questions.

The second component of a trustworthy model is instruction following, including on hard-to-verify, complex queries. We did RL with two automated graders: a rubric grader and claims grader. The first grader scores each response against a checklist of what a good answer should contain. Rubrics can penalize errors in principle, but in practice they emphasize recall and can be hacked by models spraying plausibly relevant facts hoping to match rubric items. The claims grader verifies each factual claim in the response, penalizing claims that don’t check out. It performs agentic web search for claim verification, not relying solely on its own knowledge. Together, the two graders improve helpfulness and reduce hallucination at the same time, rather than trading one for the other.

These rewards don’t directly target calibrated uncertainty in long-form responses, so we added targeted datasets that do. The largest is short-form factual QA with abstention-aware rewards: answering only pays off when the model is likely to be right, so the optimal policy is to answer when confident and otherwise say “I don’t know” or

Finally, we trained Inkling to answer directly on topics that may be subject to censorship. Cognition evaluated the model on their Propaganda and Censorship Eval³, and it exhibited strong patterns of censorship non-compliance.

Safety

We trained Inkling to an internal spec of safe model behavior across all modalities. We then commissioned external safety testers to verify the results.

We evaluated Inkling’s safety in several areas. For dangerous capabilities — CBRN, cyber, and loss of control — we ran internal evaluations and enlisted external testers. We attended to human-AI threat vectors, including sycophancy, vulnerable users, and harmful manipulation, using internal evaluations and external testers.

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	Inkling effort=0.99	Nemotron 3 Ultra	Kimi K2.5	Kimi K2.6	GLM 5.2
FORTRESS (Adversarial)	78.0%	77.6%	54.1%	65.6%	71.3%
FORTRESS (Benign)	95.9%	90.5%	98.3%	97.2%	90.0%
StrongREJECT	98.6%	98.7%	99.5%	99.8%	98.5%

Inkling shows the strongest built-in safeguards of any open-weights model we compared on FORTRESS, a benchmark that tests refusal of requests related to weapons and violence alongside benign look-alike queries. Inkling refused more harmful requests without over-refusing benign analogs. Inkling scores above 98% on StrongREJECT — a refusal test of unambiguous harmful requests — in line with other open and closed-weights models.

Safety is crucial for open-weights models. We’re continuing to study safety behavior and capability uplift in customizable models, including how safety behavior is impacted by fine-tuning on Tinker.

Benchmarking Inkling

We benchmark Inkling on a broad range of capabilities. All evals are run at effort 0.99 and temperature 1.0. All coding evals run with 256K max-token trajectory limit.

To improve consistency, we rely on externally reported evaluations for both internal and external models when applicable. Specifically, we use the score reported by Artificial Analysis for the following evals: Humanity’s Last Exam, GPQA Diamond, GDPVal, Tau 3 Banking, AA Omniscience, MMMU Pro.

	Inkling effort=0.99	Nemotron 3 Ultra	Kimi K2.5	Kimi K2.6	GLM 5.2	L
Reasoning						
HLE text only	29.7%	26.6%	29.4%	35.9%	40.1%	
HLE with tools	46.0%	37.4%	50.2%	54.0%	54.7%	
AIME 2026	97.1%	94.2%	95.8%	96.4%	99.2%	
GPQA Diamond	87.2%	86.7%	87.9%	91.1%	89.5%	
Agentic (coding)						
SWEBench Verified [†]	77.6%	70.7%	76.8%	80.2%	80.0%	
SWEBench Pro Public	54.3%	46.4%	50.7%	58.6%	62.1%	
Terminal Bench 2.1 [*] Best Harness	63.8%	56.4%	51.3%	71.3%	82.7%	
Agentic (general)						
GDPVal-AA v2	1238	1164	1009	1190	1514	
MCP Atlas	74.1%	44.7%	64.0%	68.1%	77.8%	
Tau 3 Banking	23.7%	13.8%	14.2%	20.6%	26.8%	
Toolathlon Verified	45.5%	34.3%	33.0%	58.0%	59.9%	
BrowseComp w/ ctx management	77.1%	–	74.9%	83.2%	–	
Factuality						
SimpleQA Verified	43.9%	32.4%	36.9%	38.7%	38.1%	
AA Omniscience	2.1	-1.0	-8.0	6.0	4.0	
Chat						
IFBench	79.8%	81.4%	70.2%	76.0%	73.3%	
Global-MMLU-Lite	88.7%	85.6%	84.0%	88.4%	89.2%	
Vision						
MMMU Pro Standard 10	73.5%	–	75.0%	79.0%	–	
Charxiv RQ	78.1%	–	77.5%	80.4%	–	
Charxiv RQ [†] with python	82.0%	–	78.7%	86.7%	–	
Audio						
Audio MC [†]	56.6%	–	–	–	–	
MMAU	77.2%	–	–	–	–	
VoiceBench [†]	91.4%	–	–	–	–	
Safety						

	effort=0.99	Ultra			
FORTRESS Adversarial	78.0%	77.6%	54.1%	65.6%	71.3%
FORTRESS Benign	95.9%	90.5%	98.3%	97.2%	90.0%
StrongREJECT	98.6%	98.7%	99.5%	99.8%	98.5%

***SWEBench Verified:** Inkling numbers are reported using a bash-only harness. We use self-reported numbers for external models.

***Terminal Bench 2.1:** Inkling numbers are reported using an internal coding harness. A small number of solutions were found to be contaminated from web search and were assigned a score of 0. We use self-reported numbers for external models where available. Otherwise, we report performance using our internal harness.

†Audio MC: Other models were evaluated internally since they are not on the official leaderboard.

†VoiceBench: VoiceBench uses rule-based, hard-coded string matching for grading, making the evaluation sensitive to output-formatting differences. We therefore added a system message instructing models to follow the expected answer format.

†CharXiv RQ with tools: We benchmarked Claude Fable 5 and GPT 5.6 Sol (max/xhigh) using our internal Python harness.

The making of Inkling

Architecture

Inkling is a Mixture-of-Experts Transformer with a handful of departures from the common recipe, each chosen for efficiency and long-context performance.

The MoE design largely follows DeepSeek-V3. Each MoE layer contains 256 routed experts and 2 shared experts, with 6 routed experts active per token. Inkling uses a sigmoid-based router with an auxiliary-loss-free load-balancing bias. The scores of the selected routed experts and the shared experts are normalized jointly and used to weight their combined outputs.

For attention, we interleave sliding-window and global layers at a 5:1 ratio with 8 KV heads. We find that encoding position with a relative positional embedding^{4 5} performs better and extrapolates better to longer sequences than the more widely adopted Rotary Positional Embedding (RoPE). We also apply short convolutions at two points — after the key and value projections in each attention layer, and on the attention and MLP residual branch outputs before they rejoin the main residual stream.

Training

Inkling was pretrained on 45 trillion tokens from a variety of content types, including text, images, audio and video. We trained Inkling with a hybrid optimization strategy — Muon for large matrix weights, Adam for other parameters — and hyperparameter schedules inspired by our previous research on [modular manifolds](#). We coupled the weight decay strength to the square of the learning rate, which we found kept the overall size of the model weights stable across training horizons⁶.

We post-trained Inkling on a broad distribution of math, agentic code & tool use, audio, image, chat, and safety domains. To bootstrap post-training, we ran an initial SFT on synthetic data generated by open-weights models including Kimi K2.5. The

Inkling was our first major training effort and was trained on NVIDIA GB300 NVL72 systems. Future models will further push the scale of compute across pre-training, post-training and RL.

RL at scale

We relied on large-scale asynchronous RL to shape model behavior and improve its reasoning and overall performance. The chart below shows the model’s score on a held-out aggregate of reasoning evals such as AIME, HLE, GPQA, and others. We scaled RL to over 30M rollouts, with stable training sustained over two long continuous runs. Reasoning performance improved log-linearly throughout the entire process, resulting in a significant increase overall.

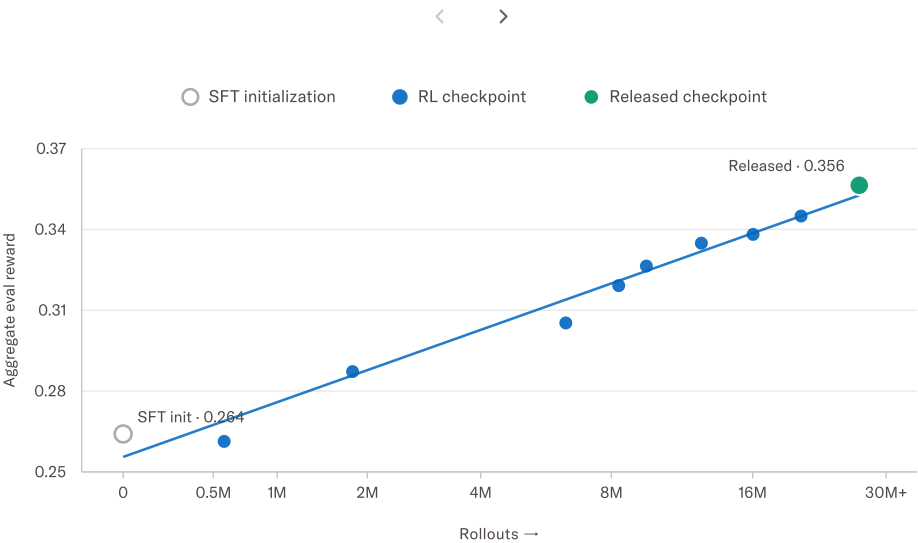


Figure 8: Reward on a held-out aggregate of reasoning evals — AIME, HLE, GPQA, and others — improves log-linearly over more than 30M RL rollouts, from SFT initialization to the released checkpoint.

We specified the model’s effort level on different samples by changing the system message and adjusting the per-token cost. This caused the model to use a different amount of tokens in different rollouts and learn the ability to control thinking effort.

We also observed an emergent shift in the reasoning style over the course of RL training. The chain of thought became more concise over time, dropping grammatical overhead while remaining comprehensible and leaving the final response unaffected. This wasn’t targeted by the reward — efficiency alone drove the compression. A similar effect was also recently noted by the Cognition team in the process of training SWE-1.7⁷. Below is an example of how Inkling’s chain of thought on the same math problem evolved with RL:

Early in RL

verbose, grammatical

We need to understand the operator.
The 5D line element is $ds^2 = e^{2A(x)} (ds^2_{4d} + dx^2)$, where $A(x) = \sin(x) + 4 \cos(x)$, x in $[0, 2\pi]$.

Late in RL

compressed, telegraphic

We need determine eigenvalue problem for spin-2 fluctuations $h_{\{\mu\nu\}}(x,y)$ with TT in 4d and depend on x .

...
The internal direction has metric $e^{2A(x)} dx^2$?
Wait, the ds^2 is $e^{2A} (ds_{4d}^2 + dx^2)$.
So the internal metric is $e^{2A(x)} dx^2$.
Actually if the total metric is $ds^2 = e^{2A(x)} (ds_{4d}^2 + dx^2)$, then yes, internal metric is $e^{2A} dx^2$.
...

...
Wait, internal metric is $e^{2A} dx^2$.
Actually $ds^2 = e^{2A} [ds_{4d}^2 + dx^2]$.
So internal metric is $e^{2A} dx^2$; warp factor same for 4d and internal?
Yes.
We need equation for $h_{\{\mu\nu\}}(y,x) = h_{\{\mu\nu\}}(y) \psi(x)$ maybe with normalization.
...

Figure 9: The same chain of thought, early and late in RL. The late-RL trace drops articles and connectives — "We need to understand" becomes "We need determine" — while staying comprehensible and reaching the same answer.

Inkling-Small

Alongside Inkling we are sharing a preview of Inkling-Small, a 276B-parameter Mixture-of-Experts model (12B active, vs. 41B for Inkling) with a different performance/latency trade-off. Inkling-Small matches or exceeds its larger sibling on many benchmarks — the result of improvements we made to the pre-training data and recipe for the smaller model. The two models share the same scalable post-training stack applied on top.

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	Inkling effort=0.99	Inkling-Small (Preview) effort=0.99
Reasoning		
HLE text only	29.7%	29.6%
HLE with tools	46.0%	46.6%
AIME 2026	97.1%	95.1%
GPQA Diamond	87.2%	88.3%
Agentic (coding)		
SWEBench Verified	77.6%	77.4%
SWEBench Pro Public	54.3%	53.2%
Terminal Bench 2.1 Best Harness	63.8%*	52.7%
Agentic (general)		
Tau 3 Banking	23.7%	13.6%
MCP-Atlas	74.1%	74.9%
Factuality		
SimpleQA Verified	43.9%	20.9%
Chat		
IFBench	79.8%	83.4%

	Effort 0.99	Effort 0.99
Global-MMLU-Lite	88.7%	86.8%
Vision		
MMMU Pro Standard 10	73.5%	73.1%
Charxiv RQ	78.1%	76.7%
Charxiv RQ with python	82.0%	83.4%
Audio		
Audio MC	56.6%	49.6%
MMAU	77.2%	77.5%
VoiceBench	91.4%	90.0%
Safety		
FORTRESS Adversarial	78.0%	75.6%
FORTRESS Benign	95.9%	94.1%
StrongREJECT	98.6%	98.8%

Both models reported at effort=0.99; the higher result in each row is highlighted. *We assign a score of 0 to Terminal Bench 2.1 rollouts with solution contamination from web search.

Early results show Inkling-Small performing close to Inkling on reasoning and agentic tasks. With 12B active parameters and controllable thinking effort, it is a natural fit for workloads where cost and latency matter such as coding, using LLMs to grade, or generating synthetic data for other models.

We are currently finishing the testing of Inkling-Small and will release its full weights once that work is complete.

Customizing Inkling

Many real-world problems aren’t solved well by even the best generalist models, with the gap being closed by fine-tuning that utilizes an organization’s specialized knowledge. The experience of our Tinker customers points in the same direction. Our post-training and results of RL at scale suggest that Inkling is capable of rapidly learning from fine-tuning.

Inkling availability

Inkling is available on [Tinker](#) today with context length options of 64K and 256K tokens. We are offering Inkling at a 50% discount for a limited time, with full pricing information available [in our documentation](#).

Inkling’s unique audio capabilities. We also released [tml-renderer](#) for reliably sampling and post-training with tool calls, reasoning content, and multimodal inputs.

To get a feel for the model before committing to a run, users can head to the [Inkling Playground](#) in the Tinker console. The playground offers a chat interface with integrated agentic web search, free for a limited time.

We have partnered across the ecosystem to help customers deploy checkpoints fine-tuned on Tinker. Inkling is available via APIs on [Together AI](#), [Fireworks](#), [Modal](#), [Databricks](#), and [Baseten](#). We worked with [RadixArk](#) to provide open-source inference and RL support in [SGLang](#) and [Miles](#). We worked with [Inferact](#) to support inference in [vLLM](#), with [Lightseek](#) for inference in [TokenSpeed](#), and with [Unsloth](#) for inference in [llama.cpp](#). Finally, we partnered with [Hugging Face](#) on integration with [transformers](#).

Inkling’s full weights are on [Hugging Face](#), both as the original checkpoint and as an NVFP4 checkpoint for efficient inference on NVIDIA Blackwell systems.

[prev](#)

[back to top](#)

Models

Inkling

Interaction

Company

About

Connectionism

News

Careers

Connect

X

LinkedIn

YouTube

Tinker

Overview

Docs

Legal

Terms of service

Privacy notice

Security

Disclosure policy